

SITKA 1 NE, AK, USA

WMO: 703730

Lat: 57.0574N

Lon: 135.3266W

Elev: 78

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 25379

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.8	18.4	4.9	7.1	19.6	9.8	9.1	22.7	4.7	42.7	4.0	42.0	0.4	N/A	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	67.6	58.9	64.7	57.7	62.2	56.6	60.3	65.1	59.2	62.8	58.1	61.1	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	73.0	60.6	57.4	70.5	60.0	56.5	68.3	59.3	26.6	65.7	25.7	62.9	25.1	61.0	66.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.5	2.9	2.8	DB	9.9	75.2	5.7	3.6	5.8	77.8	2.5	80.0	-0.7	82.0	-4.9	84.7	(4)
(5)			WB	10.1	63.0	5.4	2.1	6.2	64.5	3.1	65.8	0.0	66.9	-3.9	68.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	34.6	33.6	35.4	40.9	46.7	51.5	55.3	55.9	51.6	44.0	36.8	34.2	(6)
(7)		DBStd	9.58	6.80	6.21	5.87	3.85	3.99	3.52	3.03	2.64	3.42	4.30	6.19	6.37	(7)
(8)		HDD50	2900	478	458	454	274	118	20	1	0	22	188	396	491	(8)
(9)		HDD65	7872	942	878	919	723	568	405	300	282	403	650	846	956	(9)
(10)		CDD50	504	1	0	0	1	15	65	166	183	69	4	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	9	0	0	0	0	1	2	5	1	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	0.8	0.8	0.7	0.8	0.9	0.9	0.7	0.6	0.7	0.8	0.8	0.8	(14)	
(15)	Precipitation	PrecAvg	87.3	8.1	6.3	6.0	4.9	4.6	3.7	4.1	6.7	11.3	13.9	9.3	8.4	(15)
(16)		PrecMax	123.0	15.7	15.6	12.9	11.2	10.0	8.1	9.5	19.5	20.7	30.8	16.8	15.1	(16)
(17)		PrecMin	60.1	1.6	0.2	0.9	1.1	1.2	0.9	1.7	1.7	3.9	6.3	1.7	2.2	(17)
(18)		PrecStd	14.7	3.5	3.6	2.6	2.4	2.3	2.0	1.8	3.9	4.6	5.0	3.7	3.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	50.0	53.7	59.6	70.5	71.6	69.8	69.5	66.8	56.9	51.5	52.2	(19)
(20)			MCWB	48.6	44.3	44.7	45.5	52.3	60.1	60.9	60.7	55.9	52.4	48.0	43.9	(20)
(21)		2%	DB	47.4	47.4	49.2	54.0	61.5	64.8	66.5	66.5	62.2	54.0	49.3	47.2	(21)
(22)			MCWB	45.0	42.7	42.9	44.8	50.8	57.7	59.3	59.4	55.6	51.0	47.6	43.4	(22)
(23)		5%	DB	44.7	44.5	46.2	51.2	58.0	61.3	63.7	64.1	59.6	52.3	47.3	44.4	(23)
(24)			MCWB	42.6	41.7	41.3	44.1	50.4	55.2	57.9	58.4	55.3	49.8	45.6	42.0	(24)
(25)		10%	DB	42.7	42.3	43.6	48.7	55.0	58.6	61.0	61.7	57.7	50.6	45.2	42.1	(25)
(26)			MCWB	41.2	40.2	40.2	43.0	49.2	53.5	56.7	57.6	55.0	48.5	44.1	40.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	46.3	47.8	48.4	54.3	61.1	61.7	61.7	59.9	54.2	49.8	46.4	(27)
(28)			MDCB	53.9	47.9	54.1	52.0	62.0	70.1	68.6	66.2	61.7	55.2	50.1	47.8	(28)
(29)		2%	WB	46.2	44.1	44.7	46.3	53.0	58.4	60.1	60.3	58.0	52.6	48.1	44.2	(29)
(30)			MDCB	48.1	46.0	48.7	51.5	60.1	64.2	65.5	64.1	59.7	53.6	48.7	46.8	(30)
(31)		5%	WB	43.9	42.4	43.1	45.2	51.4	56.5	58.6	59.4	56.9	50.9	46.2	42.3	(31)
(32)			MDCB	45.4	44.3	46.2	50.2	56.6	60.5	62.5	62.5	58.7	52.0	47.1	44.3	(32)
(33)		10%	WB	42.0	40.8	41.4	44.1	50.0	54.7	57.4	58.5	55.6	49.4	44.5	40.9	(33)
(34)			MDCB	43.0	42.6	44.2	48.5	54.2	57.8	60.2	61.2	57.7	50.7	45.6	42.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	9.4	11.1	13.4	13.3	11.6	9.7	10.3	10.9	9.5	7.4	6.9	(35)
(36)			MCDBR	8.7	11.3	15.4	18.8	21.4	18.4	16.1	16.1	15.3	11.2	8.1	8.3	(36)
(37)			MCWBR	6.1	7.2	9.8	10.5	12.2	10.1	9.1	9.7	9.9	8.4	6.6	6.1	(37)
(38)		5% WB	MCDBR	8.1	8.9	13.9	14.7	16.2	15.2	13.0	12.0	12.1	8.5	7.6	7.9	(38)
(39)			MCWBR	6.8	6.5	9.5	9.3	10.1	9.1	7.9	8.2	8.7	7.4	6.8	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.312	0.331	0.395	0.372	0.369	0.381	0.360	0.354	0.339	0.304	0.280	(40)	
(41)		taud	2.249	2.385	2.374	2.189	2.384	2.441	2.447	2.527	2.503	2.398	2.272	2.189	(41)	
(42)		Ebn at Noon	181	237	263	258	274	277	270	270	254	221	176	151	(42)	
(43)		Edh at Noon	16	22	28	39	35	33	32	28	25	21	16	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	120	337	688	1155	1465	1504	1271	1097	705	374	162	86	(44)	
(45)		RadStd	18	65	99	113	157	135	150	140	119	50	24	14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	+2.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page